AP PRECALCULUS

Unit 1 • Lesson 1.6 • Notes (filled in)

Topic 1.6 • Polynomial Functions and End Behavior

Notes (filled in)

Learning goals

- Describe end behavior with limit notation.
- Determine end behavior from the degree and the sign of the leading coefficient, in standard or factored form.
- Sketch a polynomial from its zeros, multiplicities and end behavior.

Vocabulary. end behavior, limit, $\lim_{x\to\infty}$, $\lim_{x\to-\infty}$, increases without bound, leading term.

1. Limit notation

$\lim\limits_{x\to\infty} p(x) = \infty$ means: as x __increases without bound__, $p(x)$ __increases without bound__.
$\lim\limits_{x\to-\infty} p(x) = -\infty$ means: as x __decreases without bound__, $p(x)$ __decreases without bound__.

2. The leading term controls the ends

Key idea

	$a_n > 0$	$a_n < 0$
n even	left __∞__, right __∞__	left __$-\infty$__, right __$-\infty$__
n odd	left __$-\infty$__, right __∞__	left __∞__, right __$-\infty$__

Example 1

Describe the end behavior of $p(x) = 3x^2 - 2x^5 + 7$.

Leading term $-2x^5$ (highest power, not the first term written): odd degree, negative. $\lim_{x\to-\infty} p(x) = \infty$, $\lim_{x\to\infty} p(x) = -\infty$.

Example 2 • Factored form

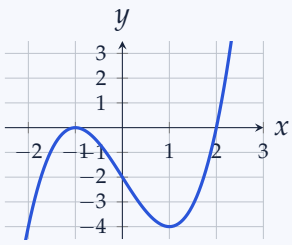
Describe the end behavior of $q(x) = -x(x-2)^2(3-x)$.

Leading term: $-x \cdot x^2 \cdot (-x) = x^4$. Even degree, positive: $\lim_{x\to\pm\infty} q(x) = \infty$.

Example 3 • Sketching

Sketch $p(x) = (x+1)^2(x-2)$ using zeros, multiplicity, y-intercept and end behavior.

Zeros: -1 (multiplicity 2, touches), 2 (crosses). y-intercept $p(0) = (1)(-2) = -2$. Leading term x^3: down on the left, up on the right.



Common mistakes

- Using the first written term as the leading term.
- In factored form, missing a negative sign hidden in a factor like $(3 - x)$.

Check your understanding

1. $\lim_{x \to -\infty}(4 - x^3) =$ ∞ .
2. Both ends go down when the degree is even and the leading coefficient is negative .